

Data Inspection Report

Based on the exploratory data analysis, the student population displays specific distributions that must be accounted for to prevent algorithmic bias. Examining the target attribute revealed a high baseline dropout rate of approximately 33%. This significant rate highlights the importance of an intervention to improve student well-being and mitigate financial losses for the institution.

Demographically, the institution shows a gender imbalance, consisting of nearly 65% females and 35% males, with a median age of 20 years. The student body is highly homogenous: 97.5% originate from Portugal, with the remainder primarily from other Portuguese-speaking countries. Finally, the most prevalent student background was secondary school.

An initial risk assessment by factor revealed distinct patterns. While international status and lack of a scholarship hovered near the baseline dropout rate, male gender was correlated with a significantly higher dropout rate of nearly 50%. Notably, financial factors were the strongest predictors: being a debtor (62%) and having unpaid tuition (86.6%) were highly correlated with attrition. This is likely driven by Portuguese legislation, which prohibits students with unpaid tuition from taking exams, effectively forcing them to drop out.

Regarding sensitive attributes in the data, we classified the variables into two categories:

1. Personally Sensitive Attributes

This category includes nationality, international status, and displaced status. These variables serve as proxies for race, ethnicity, and migration background; failing to address them risks introducing xenophobia and bias against minority groups. Furthermore, "Educational special needs" acts as a proxy for medical and/or learning disabilities, carrying a risk of ableist discrimination. Lastly, marital status, gender, and age constitute highly personal information where bias could lead to discrimination, sexism, and ageism.

2. Socio-Economic Sensitive Attributes

This category consists of debtor status, tuition fee status, and scholarship holdings. All three are direct proxies for a student's financial situation, and ignoring these risks baking in prejudice against lower-income students. Finally, the mother's and father's qualifications and occupations serve as proxies for social class and familial wealth, which could introduce significant class bias into the model.